

Flutter Roadmap

Stage 02 – Problem solving

Prepared by:

Momen Mohammed Kraizem

Computer Engineering Graduate

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Practical Exercise 1

Even or Odd Number

Problem:

Determine whether a given number is even or odd

Input	process	Output
Integer number	Check whether the remainder of dividing the number by 2 equals zero	Even or Odd

Algorithm:

1. Start.
2. Read the number.
3. If the remainder of dividing the number by 2 equals 0, display "Even".
4. Otherwise, display "Odd".
5. End.

Pseudocode

```
BEGIN
INPUT number
IF number MOD 2 = 0 THEN
    OUTPUT "Even"
ELSE
    OUTPUT "Odd"
END IF
END
```

Practical Exercise 2

Rectangle Area

Problem:

Calculate the area of a rectangle.

Input	process	Output
Length, Width	Multiply the length by the width	Rectangle area

Algorithm:

1. Start.
2. Read the length.
3. Read the width.
4. Calculate the area.
5. Display the area.
6. End.

Pseudocode

BEGIN

INPUT length

INPUT width

area = length * width

OUTPUT area

END

Practical Exercise 3

Student Average

Problem:

Calculate the average of three student grades.

Input	process	Output
Grade 1, Grade 2, Grade 3	Add the three grades and divide the result by 3	Student average

Algorithm:

1. Start.
2. Read the first grade.
3. Read the second grade.
4. Read the third grade.
5. Calculate the average.
6. Display the average.
7. End.

Pseudocode

BEGIN

INPUT grade1

INPUT grade2

INPUT grade3

average = (grade1 + grade2 + grade3) / 3

OUTPUT average

END

Practical Exercise 4

Largest of Two Numbers

Problem:

Determine the larger of two numbers.

Input	process	Output
First number, Second number	Compare the two numbers.	Largest number

Algorithm:

1. Start.
2. Read the first number.
3. Read the second number.
4. Compare the two numbers.
5. Display the larger number.
6. End.

Pseudocode

```
BEGIN
```

```
INPUT number1
```

```
INPUT number2
```

```
IF number1 > number2 THEN
```

```
    OUTPUT number1
```

```
ELSE
```

```
    OUTPUT number2
```

```
END IF
```

```
END
```

Practical Exercise 5

Celsius to Fahrenheit

Problem:

Convert a temperature from Celsius to Fahrenheit.

Input	process	Output
Temperature in Celsius	Convert Celsius to Fahrenheit using the conversion formula.	Temperature in Fahrenheit

Algorithm:

1. Start.
2. Read the temperature in Celsius.
3. Convert the temperature to Fahrenheit.
4. Display the Fahrenheit temperature.
5. End.

Pseudocode

BEGIN

INPUT Celsius

Fahrenheit = (Celsius * 9 / 5) + 32

OUTPUT Fahrenheit

END